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### BATTERY DEVICE

#### Abstract

A battery device includes: battery modules; and a housing configured to accommodate the battery modules. The housing configured to accommodate a pair of strip-shaped bus bars and a member retaining the pair of bus bars, the member includes a retainer with one side open and having a U-shaped cross section and first and a second cover portions that are rotatably disposed on the retainer, a first bus bar of the pair of bus bars is accommodated between the retainer and the first cover portion, a second bus bar of the pair of bus bars is accommodated between the first cover portion and the second cover portion, and a pressed portion is formed on the first cover portion, and is formed to be pressed and cause the first cover portion to rotate when the first bus bar is placed between the retainer and the first cover portion.

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## Background/Summary

### CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is based on and claims priority under 35 USC 119 from Chinese Patent Application No. 20/241,0268234.8 filed on Mar. 9, 2024, the contents of which are incorporated herein by reference.

### TECHNICAL FIELD

[0002] The disclosure relates to a battery device loaded in an electric vehicle and the like.

### BACKGROUND

[0003] In recent years, with increasing attention to environmental protection and sustainable development, secondary batteries, particularly applications in the field of electric vehicles, have attracted great attention. These batteries are not only affordable and reliable, but also must be sustainable and advanced to ensure efficient utilization of energy. The popularity of electric vehicles is becoming an important trend in the development of modern transportation, and its core relies on an efficient and long-life secondary battery technology. The development and improvement of this technology is critical for reducing dependence on fossil fuels, lowering greenhouse gas emissions, and promoting energy transfer. Therefore, research and development of the secondary batteries are not only important aspects of technical innovation but also a key to a cleaner and greener transportation system.

[0004] In the electric vehicles, a battery device (also referred to as a “battery pack”) is one of core components responsible for powering the vehicle. The battery pack consists of the plurality of secondary battery modules accommodated in a battery pack housing. Integration and management of the battery modules through the battery pack is important for improving the range of the electric vehicle, reducing charging time and improving overall energy efficiency. See Japanese Patent Application Laid-Open Publication No. 2021125430A.

[0005] A battery pack in an electric vehicle is an important component and is formed by combining a plurality of battery modules, and the battery modules are connected to each other through bus bars to form a unified energy supply system.

[0006] A main function of these bus bars is to connect battery cells and connect the battery pack to a main switch and a junction box of the vehicle. The arrangement of the bus bars needs to be considered in the design of battery packs since a layout thereof affects an overall size and an efficiency of the battery pack. If the arrangement of the bus bars is unreasonable, it will cause the battery pack to increase in volume and occupy too much cabin space, and may affect the performance and reliability of the battery pack.

[0007] Due to a limited interior space of the vehicle, it is necessary to find a suitable balance between sufficient battery capacity and the interior space of the vehicle. In practical applications, in order to fix and manage the bus bars in a narrow battery pack housing space, fixing members such as clips are usually used. In the process of manufacturing and assembling the battery pack, simplifying the number of the fixing members not only reduces material costs, but also helps reduce assembly complexity, thereby improving production efficiency and reducing overall costs.

[0008] The present disclosure provides a battery device capable of properly disposing bus bars and reducing the number of fixing members.

### SUMMARY

[0009] According to a first aspect of the present disclosure, there is provided a battery device for an electric vehicle including: [0010] a plurality of battery modules; and [0011] a housing configured

to accommodate the plurality of battery modules, in which [0012] the housing is configured to accommodate a pair of strip-shaped bus bars and a retaining member retaining the pair of bus bars, [0013] the retaining member includes [0014] a retaining portion with one side open and having a U-shaped cross section, and [0015] a first cover portion and a second cover portion that are rotatably disposed on the retaining portion, [0016] a first bus bar of the pair of bus bars is accommodated between the retaining portion and the first cover portion, [0017] a second bus bar of the pair of bus bars is accommodated between the first cover portion and the second cover portion, and [0018] a pressed portion is formed on the first cover portion, and is formed to be pressed and cause the first cover portion to rotate when the first bus bar is placed between the retaining portion and the first cover portion.

[0019] According to a second aspect of the present disclosure, there is provided a method for assembling bus bars in a battery device for an electric vehicle using a retaining member, in which [0020] the battery device includes a plurality of battery modules and a housing configured to accommodate the plurality of battery modules, [0021] the housing is configured to accommodate the pair of strip-shaped bus bars and the retaining member retaining the pair of bus bars, [0022] the retaining member includes a retaining portion with one side open and having a U-shaped cross section, and a first cover portion and a second cover portion that are rotatably disposed on the retaining portion, [0023] a pressed portion is formed on the first cover portion, and [0024] the method includes: [0025] pressing a first bus bar of the pair of bus bars toward the pressed portion to rotate the first cover portion, and accommodating the first bus bar between the retaining portion and the first cover portion; [0026] disposing a second bus bar of the pair of bus bars on the rotated first cover portion; and [0027] rotating the second cover portion to accommodate the second bus bar between the first cover portion and the second cover portion.

[0028] According to the battery device in the present disclosure, the two bus bars can be fixed by using one retaining member, the requirement of two or more independent retaining members in the past is eliminated, the number of parts is reduced, and operation steps of installing the bus bars are simplified. Moreover, a space can be efficiently utilized in the compact arrangement of the battery modules, and a space layout in the battery pack can be optimized.

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## Description

### BRIEF DESCRIPTION OF DRAWINGS

[0029] Exemplary embodiment(s) of the present disclosure will be described in detail based on the following figures, wherein:

[0030] FIG. 1 is a plan view of a battery pack according to an embodiment of the present disclosure;

[0031] FIG. 2 is a perspective view of a retaining member according to the first embodiment of the present disclosure;

[0032] FIG. 3 is a schematic side view of the retaining member in an open state according to the first embodiment of the present disclosure;

[0033] FIG. 4 is a schematic side view of the retaining member in a closed state according to the first embodiment of the present disclosure;

[0034] FIGS. 5A to 5D are schematic diagrams showing steps of installing bus bars in the retaining member according to the first embodiment of the present disclosure;

[0035] FIG. 6 is a perspective view of a retaining member according to a second embodiment of the present disclosure;

[0036] FIG. 7 is a schematic side view of the retaining member in an open state according to the second embodiment of the present disclosure;

[0037] FIG. 8 is a schematic side view of the retaining member in a closed state according to the

second embodiment of the present disclosure;

[0038] FIGS. **9A** to **9D** are schematic diagrams showing steps of installing bus bars in the retaining member according to the second embodiment of the present disclosure;

[0039] FIG. **10** is a perspective view of a retaining member according to a third embodiment of the present disclosure;

[0040] FIG. **11** is a schematic side view of the retaining member in an open state according to the third embodiment of the present disclosure;

[0041] FIG. **12** is a schematic side view of the retaining member in a closed state according to the third embodiment of the present disclosure; and

[0042] FIGS. **13A** to **13D** are schematic diagrams showing steps of installing bus bars in the retaining member according to the third embodiment of the present disclosure.

## DESCRIPTION OF EMBODIMENTS

[0043] Hereinafter, embodiments of the present disclosure will be described with reference to the accompanying drawings. The same structures in the drawings are denoted by the same reference numerals.

### First Embodiment

#### Battery Pack

[0044] FIG. **1** is a plan view of a battery pack according to an embodiment of the present disclosure, in which shaded boxes represent battery modules **11**. As shown in FIG. **1**, a battery pack **1** of the present disclosure includes the plurality of battery modules **11** (twelve in the present embodiment) and a housing **12** accommodating these battery modules **11**, and can be mounted in an electric vehicle such as a pure electric vehicle, a hybrid electric vehicle, a fuel cell electric vehicle and the like. Each battery module **11** is formed by stacking a plurality of battery cells in a left-right direction.

[0045] In the housing **12**, the battery modules **11** are arranged in two rows, each row including six battery modules. One end of the housing **12** is provided with a first junction box **13**, and the other end of the housing **12** is provided with a second junction box **14**. The first junction box **13** and the second junction box **14** are used for supplying electric power from the battery modules **11** to a motor of the vehicle, and supplying electric power received from the motor to the battery **11**. The housing **12** containing the battery modules **11**, the first junction box **13**, and the second junction box **14** is closed by a housing cover (not shown).

[0046] The first junction box **13** and the second junction box **14** are electrically connected through bus bars **20**. The bus bar is usually made of a highly conductive material and is covered with insulating material on an outer periphery thereof. The bus bar is a connecting member inside the battery pack in the electric vehicle, a main function is to collect and distribute current, and in a traveling process of the electric vehicle, a large amount of current is required to drive the motor, and these currents are provided by the battery modules and transmitted through the bus bars. When the driver depresses a gas pedal or a brake pedal, the battery pack outputs or recovers the current according to the demand, and the bus bars ensure that these currents can quickly and accurately reach a designated location.

[0047] In the present embodiment, the first junction box **13** and the second junction box **14** are electrically connected through two bus bars **20a** and **20b** (also collectively referred to as “bus bars **20**” in the specification). The bus bars **20a** and **20b** both have a strip shape, and have substantially the same thickness (a shortest side among three sides of the strip-shaped bus bar) and the same width (a second shortest side among the three sides of the strip-shaped bus bar). One end of each of the bus bars **20a** and **20b** is connected to a junction terminal of the first junction box **13** and extends laterally from the first junction box **13**, and the bus bars **20a** and **20b** are stacked up and down in a horizontal state. Here, the “horizontal state” and the “laterally extending” of the bus bars refer to that a length direction and a width direction of the strip-shaped bus bars **20** are substantially in a plane parallel to a bottom surface of the housing **12** (a paper surface direction of FIG. **1**), and a

thickness direction is generally along a direction perpendicular to the bottom surface of the housing **12** (a direction perpendicular to the paper surface in FIG. **1**). The two bus bars **20a** and **20b** stacked up and down in the horizontal state are manually bent and then connected to the second junction box **14** through the battery modules to enable electrical connection between the first junction box **13** and the second junction box **14**. The layout of the bus bars includes a step of bending the bus bars **20a** and **20b** that are stacked up and down in the horizontal state directly downward and then along a length direction, so that the broadening of the bus bars **20a** and **20b** are widened in a vertical direction, that is, the two bus bars **20a** and **20b** extend longitudinally side by side in an upright state. Here, the “upright state” and “longitudinally extending” refer to that the length direction and the thickness direction of the strip-shaped bus bars **20** are substantially in a plane parallel to the bottom surface of the housing **12** (the paper surface direction of FIG. **1**), and the width direction is generally along the direction perpendicular to the bottom surface of the housing **12** (the direction perpendicular to the paper surface in FIG. **1**). In this way, the two bus bars **20a** and **20b** can be effectively disposed side by side between the closely arranged battery modules to adapt to space limitation inside the battery pack and optimize space utilization in the battery pack. [0048] In order to reduce displacement of the bus bars **20** caused by vibration or movement, improve the stability and reliability of a battery pack device, as shown in FIG. **1**, a plurality of retaining member **30a** to **30d** for fixing the bus bars **20** are mounted on the housing **12**. One end of each of the bus bars **20a** and **20b** is connected to the junction terminal of the first junction box **13** and extends laterally from the first junction box **13**, and the bus bars **20a** and **20b** are stacked up and down in the horizontal state, and in the present embodiment, the bus bar **20a** is located above the bus bar **20b**. The bus bars **20a** and **20b** stacked up and down in the horizontal state are fixed in the retaining member **30a**, then pass through the battery modules side by side in the upright state after being bent, are sequentially fixed in the retaining members **30b**, **30c**, and **30d**, and are finally connected to the second junction box **14**.

[0049] The retaining member **30** of the present disclosure can retain the two bus bars **20a** and **20b** with one retaining member **30**, and thus the number of retaining members **30** is reduced. The structure of the retaining member **30** in the housing **12** of the battery pack **1** and retention of the bus bars **20a** and **20b** by the retaining member **30** will be described below with reference to the accompanying drawings. In the drawings, the front of the retaining member is denoted as Fr, the rear is denoted as Rr, a left side is denoted as L, a right side is denoted as R, an upper side is denoted as U, and a lower side is denoted as D. It should be noted that the front, rear, left, right, upper, and lower sides of the retaining member have nothing to do with the front, rear, left, right, upper, and lower sides of the battery pack on which the retaining member is mounted and the vehicle.

#### Retaining Member

[0050] FIG. **2** is a perspective view of the retaining member **30** according to the first embodiment of the present disclosure, FIG. **3** is a schematic side view of the retaining member **30** in an open state according to the first embodiment of the present disclosure, and FIG. **4** is a schematic side view of the retaining member **30** in a closed state according to the first embodiment of the present disclosure. As shown in the drawings, the retaining member **30** includes a retaining portion **31** having a U-shaped cross section, and a first cover portion **32** and a second cover portion **33** rotatably fixed to the retaining portion **31**.

[0051] Specifically, in the retaining member **30** of the present embodiment, the retaining portion **31** is composed of a first sidewall portion **311**, a second sidewall portion **312**, a bottom surface portion **313** connecting them, and a mounting portion **314** located at the end where the first sidewall portion **311** of the bottom surface portion **313** is located and extending downward from the bottom surface portion **313**. The first sidewall portion **311**, the second sidewall portion **312**, the bottom surface portion **313** and the mounting portion **314** are integrally formed and form a flat-bottomed U shape opened upward in a side view, and the height of the first sidewall portion **311** is greater than

the height of the second sidewall portion **312**. Two rotating shafts **316** are formed on left and right sides of an upper end of the first sidewall portion **311**, extend from a left end and a right end respectively to the center by a predetermined length, and are used for cooperating with pivot hole portions **335** of the second cover portion **33** to be described below to mount the second cover portion **33**. Two rotating shafts **315** are formed on left and right sides of an upper end of the second sidewall portion **312**, and extend from a left end and a right end respectively to the center by a predetermined length for cooperating with pivot hole portions **325** of the first cover portion **32** to be described below to mount the first cover portion **32**. The mounting portion **314** is formed in a socket shape that cooperates with a mounting slot of the housing **12** and configured to mount the retaining member **30** in the housing **12**. In order to meet requirements of different installing directions of the retaining member **30**, the mounting portion **314** may adopt a rotatable structure or have a socket in two directions perpendicular to each other.

[0052] The first cover portion **32** includes a flat plate-shaped cover plate portion **321**, an engagement portion **324** extending in a first direction perpendicular to a plane of the cover plate portion **321** and formed at one end of the cover plate portion **321** in a front-rear direction, a pressed portion **322** extending in the first direction and formed at the other end of the cover plate portion **321** in the front-rear direction, a contact wall **323** extending in a second direction opposite to the first direction, and the two pivot hole portions **325** extending obliquely from the cover plate portion and formed on left and right sides of the pressed portion **322**. The pivot hole portion **325** is formed with a pivot hole that cooperates with the rotating shaft **315** of the retaining portion **31**, and the first cover portion **32** can be rotatably mounted on the retaining portion **31** by sleeve-fitting the pivot hole onto the rotating shaft **315**. A front end of the engagement portion **324** is formed with a hook-shaped engaging portion (not shown), and the engaging portion can be engaged with a slot (not shown) formed on an outside of the first sidewall portion **311** of the retaining portion **31**, so that the first cover portion **32** can be fixed to the retaining portion **31**.

[0053] The second cover portion **33** includes a flat plate-shaped cover surface portion **331**, and a sidewall portion **332** and an engagement portion **333** extending in a direction perpendicular to the plane of the cover surface portion **331** toward the same side formed at two ends of the cover surface portion **331** in the front-rear direction, respectively. The two pivot hole portions **335** extending obliquely from the sidewall portion **332** are formed at both ends of an outside of the sidewall portion **332**, respectively. The pivot hole portion **335** is formed with a pivot hole that cooperates with the rotating shaft **316** of the retaining portion **31**, and the second cover portion **33** can be rotatably mounted on the retaining portion **31** by sleeve-fitting the pivot hole onto the rotating shaft **316**. The front end of the engagement portion **333** extends to form a hook-shaped engaging portion **334** in a side view, and the engaging portion **334** can be engaged with a slot (not shown) formed on an outside of the second sidewall portion **312** of the retaining portion **31**, so that the second cover portion **33** can be fixed to the retaining portion **31**.

[0054] The first cover portion **32** and the second cover portion **33** are each rotatably mounted on the retaining portion **31**, and the retaining portion **31**, the first cover portion **32**, and the second cover portion **33** are sequentially arranged from bottom to top. As shown in FIG. 4, when the first cover portion **32** is turned to the closed state, that is, the cover plate portion **321** is substantially parallel to the bottom surface portion **313** of the retaining portion **31**, surfaces of the cover plate portion **321** and the bottom surface portion **313** of the retaining portion **31** facing each other, an inner side surface of the first sidewall portion **311** of the retaining portion **31**, and a surface of the pressed portion **322** of the first cover portion **32** facing the inner side surface constitute a first accommodating space **34** for accommodating one of the bus bars **20a** and **20b**. When the second cover portion **33** is also turned to the closed state, that is, when the cover surface portion **331** is substantially parallel to the cover plate portion **321** of the first cover portion **32**, surfaces of the cover surface portion **331** of the second cover portion **33** and the cover plate portion **321** of the first cover portion **32** facing each other, an inner side surface of the sidewall portion **332** of the second

cover portion **33**, and a surface of the contact wall **323** of the first cover portion **32** facing the inner side surface constitute the second accommodating space **35** for accommodating the other of the bus bars **20a** and **20b**.

[0055] FIGS. **5A** to **5D** are schematic diagrams showing steps of installing the bus bars in the retaining member according to the first embodiment of the present disclosure. The steps of installing the bus bars **20a** and **20b** in the retaining member **30** are described in detail below with reference to FIGS. **5A** to **5D**. It should be noted that the following description is based on the installing of two bus bars **20a** and **20b** stacked up and down in the retaining member **30** (that is, the retaining member **30a** in FIG. **1**) as an example. For the installing of two bus bars **20a** and **20b** disposed side by side in the retaining member **30** (that is, the retaining members **30b** to **30d** in FIG. **1**), it is sufficient to rotate each member and the operating direction by **90** degrees in the following steps.

[0056] First, as shown in FIG. **5A**, in the retaining member **30**, the first cover portion **32** and the second cover portion **33** are opened. At this stage, the installer starts to install a first bus bar (taking the bus bar **20a** as an example). The installer presses the bus bar **20a** into the retaining member **30** from above, as indicated by the arrow. During this process, the bus bar **20a** comes into contact with and pushes the pressed portion **322** of the first cover portion **32** when pressed downward, which causes the first cover portion **32** to rotate downward about an axis (the rotating shaft **315**) and enter a closed state, as shown in FIG. **5B**. As the bus bar **20a** continues moving until the bus bar **20a** comes into contact with the bottom surface portion **313** of the retaining portion **31**, the pressed portion **322** of the first cover portion **32** is pushed into contact with the inner side surface of the first sidewall portion **311** of the retaining portion **31**. At this time, the first cover portion **32** is completely closed, and the bus bar **20a** is stably accommodated in the first accommodating space **34**. Then, the installer presses the first cover portion **32** downward to engage the engaging portion formed at the front end of the engagement portion **324** with the slot formed on the outside of the first sidewall portion **311** of the retaining portion **31**, thereby fixing the first cover portion **32** to the retaining portion **31** to complete installation of the bus bar **20a**.

[0057] Then, as shown in FIG. **5C**, the installer places the second bus bar (bus bar **20b**) from above into the second accommodating space **35** in a similar way. In the last step, as shown in FIG. **5D**, the installer pushes the second cover portion **33** to the closed state and presses the second cover portion **33** downward, so that the engaging portion **334** of the second cover portion **33** is engaged with the slot formed on the outside of the second sidewall portion **312** of the retaining portion **31**, thereby fixing the second cover portion **33** to the retaining portion **31**, and completing the installation of the bus bar **20b**.

#### Assembly Steps

[0058] Steps of assembling the retaining member **30** and the bus bars **20a** and **20b** within the housing **12** of the battery pack **1** are described below.

[0059] After the battery cells and the first and second junction boxes are mounted in the housing **12**, the retaining member **30** is mounted at a predetermined position. As shown in FIG. **1**, four retaining members **30a** to **30d** are provided as an example of the present disclosure, the retaining member **30a** is for retaining the bus bars **20a** and **20b** stacked up and down in the horizontal state, and the retaining members **30b** to **30d** retain the bus bars **20a** and **20b** disposed side by side in the upright state.

[0060] Then, the first bus bar (taking the bus bar **20a** as an example) is firstly installed, and two ends of the bus bar **20a** are connected to the first and second junction boxes, respectively. Then, the bus bar **20a** is sequentially installed into the retaining members **30a** to **30d**, that is, the bus bar **20a** is properly positioned in the first accommodating space **34**, and the first cover portion **32** falls to be substantially parallel to the bottom surface portion **313** of the retaining portion **31**. Finally, two terminals of the bus bar **20a** are fixed by bolts.

[0061] Next, a second bus bar (bus bar **20b**) is installed, two ends of the bus bar **20b** are connected

to the first and second junction boxes, respectively, and then the bus bar **20b** is sequentially installed into the retaining members **30a** to **30d**, that is, the bus bar **20b** is properly located in the second accommodating space **35**, and finally the two terminals of the bus bar **20b** are fixed by bolts.

[0062] Finally, the second cover portion of each of the retaining members **30a** to **30d** is closed and gently pressed to engage the engagement portion of the upper cover with the retaining portion to complete the installation.

[0063] In this way, compared with the a method in the related art, the retaining member **30** in the first embodiment of the present disclosure allows the two bus bars **20a** and **20b** to be fixed in the same retaining member, so that the need for more than two independent retaining members is eliminated. By allowing the two bus bars **20a** and **20b** to be stacked up and down or disposed side by side, space can be efficiently utilized in the compact arrangement of the battery modules, and a space layout in the battery pack can be optimized. Moreover, by adopting the design of the pressed portion **322**, the retaining member **30** automatically pushes the first cover portion **32** to rotate and close when installing the first bus bar. This design not only reduces the number of parts, but also simplifies the operation steps of installing the bus bars. In this way, the present disclosure not only improves assembly efficiency, but also enhances overall structural stability and a space utilization rate of the battery pack.

## Second Embodiment

### Retaining Member

[0064] A retaining member **30A** according to a second embodiment of the present disclosure will be described below with reference to the drawings. In the following description, description of the same components as those of the retaining member **30** of the first embodiment will be omitted or simplified. In the retaining member **30** of the first embodiment, the first cover portion **32** and the second cover portion **33** are respectively rotatably fixed to the first sidewall portion **311** and the second sidewall portion **312** of the retaining portion **31**, respectively, while in the retaining member **30A** of the second embodiment, the first cover portion and the second cover portion are rotatably fixed to the same sidewall of the retaining portion.

[0065] FIG. **6** is a perspective view of the retaining member **30A** according to the second embodiment, FIG. **7** is a schematic side view of the retaining member **30A** in an open state according to the second embodiment of the present disclosure, and FIG. **8** is a schematic side view of the retaining member **30A** in a closed state according to the second embodiment. As shown in the drawings, the retaining member **30A** includes the retaining portion **31** having a U-shaped cross section, and the first cover portion **32** and the second cover portion **33** rotatably fixed to the retaining portion **31**.

[0066] Specifically, in the retaining member **30A** of the present embodiment, the retaining portion **31** is composed of the first sidewall portion **311**, the second sidewall portion **312**, the bottom surface portion **313** connecting them, and the mounting portion **314** located at the end where the first sidewall portion **311** of the bottom surface portion **313** is located and extending downward from the bottom surface portion **313**. The first sidewall portion **311**, the second sidewall portion **312**, the bottom surface portion **313** and the mounting portion **314** are integrally formed and form a flat-bottomed U shape opened upward in a side view. The first sidewall portion **311** is formed with two rotating shaft portions **319a** extending inward in a direction perpendicular to an inner side surface thereof, and the two rotating shaft portions **319a** respectively have rotating shafts **315** extending from two ends to the center by a predetermined length for cooperating with the pivot hole portions **325** of the first cover portion **32** to be described below to mount the first cover portion **32**. An upper end of the first sidewall portion **311** is formed with two rotating shaft portions **319b** extending outward in a direction perpendicular to an outer side surface thereof, and the two rotating shaft portions **319b** respectively have rotating shafts **316** extending from two ends to the center by a predetermined length for cooperating with the pivot hole portions **335** of the second

cover portion **33** to be described below to mount the second cover portion **33**. As shown in FIG. 7, the rotating shaft portion **319a** is formed at a certain distance from a left end and a right end of the first sidewall portion **311**, and the rotating shaft portion **319b** is formed at a position closer to the left end and the right end than the rotating shaft portion **319a**. The outside of the second sidewall portion **312** is formed with an engagement portion **317**, and an upper end of the engagement portion **317** extends to form an engaging portion **318** with a hook-shaped side surface in a side view.

[0067] The first cover portion **32** includes the flat plate-shaped cover plate portion **321**, the pressed portion **322** extending in the first direction perpendicular to the plane of the cover plate portion **321** and formed at one end of the cover plate portion **321** in the front-rear direction, and the contact wall **323** extending in the second direction opposite to the first direction and formed at the other end of the cover plate portion **321** in the front-rear direction. The contact wall **323** extends in the first direction to form a fixing portion **323a** protruding outward. The two pivot hole portions **325** extending outward are formed at a left end and a right end of a connection point of the cover plate portion **321** and the pressed portion **322**. The pivot hole portion **325** is formed with the pivot hole that cooperates with the rotating shaft **315** of the retaining portion **31**, and the first cover portion **32** is rotatably mounted on the retaining portion **31** by sleeve-fitting the pivot hole onto the rotating shaft **315**.

[0068] The second cover portion **33** includes the flat cover surface portion **331**, and the sidewall portion **332** extending in the direction perpendicular to the plane of the cover surface portion **331** and formed at one end of the cover surface portion **331** in the front-rear direction. The other end of the cover surface portion **331** in the front-rear direction is formed with the two pivot hole portions **335** extending obliquely from the cover surface portion **331**. The pivot hole portion **335** is formed with a pivot hole that cooperates with the rotating shaft **316** of the retaining portion **31**, and the second cover portion **33** is rotatably mounted on the retaining portion **31** by sleeve-fitting the pivot hole onto the rotating shaft **316**. An outside of the engagement portion **333** extends outward to form a protruding portion **333a**, and the second cover portion **33** can be fixed to the retaining portion **31** by engaging the engaging portion **318** of the retaining portion **31** with the protruding portion **333a**. Further, as shown in FIG. 8, at this time, a lower end of the engagement portion **333** abuts on the fixing portion **323a** of the first cover portion **32**, so that the first cover portion **32** is also fixed. A protrusion **336** protruding from the cover surface portion **331** is formed on an inner side of the pivot hole portion **335**. In this way, a groove **337** for accommodating the bus bars **20** is formed between an inner side surface of the cover surface portion **331**, the protrusion **336**, and the engagement portion **333**.

[0069] The first cover portion **32** and the second cover portion **33** are each rotatably mounted on the retaining portion **31**, and the retaining portion **31**, the first cover portion **32**, and the second cover portion **33** are sequentially arranged from bottom to top. As shown in FIG. 8, when the first cover portion **32** is turned to the closed state, that is, the cover plate portion **321** is substantially parallel to the bottom surface portion **313** of the retaining portion **31**, surfaces of the cover plate portion **321** and the bottom surface portion **313** of the retaining portion **31** facing each other, an inner side surface of the second sidewall portion **312** of the retaining portion **31**, and a surface of the pressed portion **322** of the first cover portion **32** facing the inner side surface constitute the first accommodating space **34** for accommodating one of the bus bars **20a** and **20b**. When the second cover portion **33** is also turned to the closed state, that is, when the cover surface portion **331** is substantially parallel to the cover plate portion **321** of the first cover portion **32**, the groove **337** of the second cover portion **33**, a surface of the cover plate portion **321** of the first cover portion **32** facing the groove **337**, and an inner surface of the contact wall **323** of the first cover portion **32** constitute the second accommodating space **35** for accommodating the other of the bus bars **20a** and **20b**.

[0070] FIGS. 9A to 9D are schematic diagrams showing steps of installing the bus bars in the

retaining member according to the second embodiment of the present disclosure. The steps of installing bus bars **20a** and **20b** in the retaining member **30A** are described in detail below with reference to FIGS. **9A** to **9D**. It should be noted that the following description is based on the installing of two bus bars **20a** and **20b** stacked up and down in the retaining member **30A** as an example. For the installing of two bus bars **20a** and **20b** disposed side by side in the retaining member **30A**, it is sufficient to rotate each member and the operating direction by 90 degrees in the following steps.

[0071] As shown in FIG. **9A**, in the retaining member **30A**, the first cover portion **32** and the second cover portion **33** are opened. At this point, the installer can start an installation process. The installer selects one bus bar (taking the bus bar **20a** as an example) and presses it from above into the retaining member **30A**, as indicated by an arrow. During this process, the bus bar **20a** comes into contact with and pushes the pressed portion **322** of the first cover portion **32** when pressed downward. This action triggers the first cover portion **32** to rotate downward about an axis (rotating shafts **315**) into the closed state, as shown in FIG. **9B**. As the bus bar **20a** continues moving until the bus bar **20a** comes into contact with the bottom surface portion **313** of the retaining portion **31**, the pressed portion **322** of the first cover portion **32** is pushed into contact with the inner side surface of the first sidewall portion **311** of the retaining portion **31**. At this time, the first cover portion **32** is completely closed and covers the bus bar **20a**, and the bus bar **20a** is stably accommodated in the first accommodating space **34**, thereby completing installation of the bus bar **20a**.

[0072] Then, as shown in FIG. **9C**, the installer places the second bus bar (bus bar **20b**) from above into the second accommodating space **35** in a similar way. In the final step, as shown in FIG. **9D**, the installer pushes the second cover portion **33** so as to rotate the second cover portion **33** to the closed state and presses the second cover portion **33** downward, so that the protruding portion **333a** of the second cover portion **33** is engaged with the engaging portion **318** of the retaining portion **31**, thereby fixing the second cover portion **33** to the retaining portion **31**. At this time, as shown in FIG. **9D**, the lower end of the engagement portion **333** of the second cover portion **33** also abuts on the fixing portion **324** of the first cover portion **32**, so that the first cover portion **32** is also fixed. In this manner, the installation of the bus bars **20a** and **20b** is completed.

[0073] Similar to the retaining member **30** of the first embodiment, the retaining member **30A** in the second embodiment of the present disclosure allows the two bus bars **20a** and **20b** to be fixed in the same retaining member, so that the need for two or more independent retaining members in the past is eliminated, space can be efficiently utilized in the compact arrangement of the battery modules, and a space layout in the battery pack can be optimized. Moreover, by using the design of the pressed portion **322**, the first cover portion **32** is automatically pushed to rotate and close when installing the first bus bar, the number of parts is reduced, and the operation steps of installing the bus bars are also simplified.

[0074] In addition, in the second embodiment of the present disclosure, the design of the retaining member **30A** allows the installer to engage the retaining portion **31** and fix the first cover portion **32** and the second cover portion **33** together by pressing only the second cover portion **33** without pressing the first cover portion **32** and the second cover portion **33** in sequence. This design further simplifies the operation steps, improves installation efficiency, and makes it easier for the bus bars **20** to be fixed in the retaining member **30A**. Moreover, since the first cover portion **32** and the second cover portion **33** are opened toward the same side and an opening degree is relatively large, the operation space is larger and it is easier to visually recognize the bus bars **20** when the bus bars **20** are installed.

[0075] In addition, the second cover portion **33** is provided with the protrusion **336** on an inner side surface of the cover surface portion **331**, the groove **337** for accommodating the bus bars **20** is formed between the protrusion **336** and the engagement portion **333**, the groove **337** and the inner side surface of the contact wall **323** constitute the second accommodating space **35**, limits the

displacement of the bus bars in the width direction, and thus effectively limits the unnecessary displacement of the bus bars in the width direction. Accordingly, potential damage caused by vibration or movement is reduced, and an additional guarantee is added to the stability and reliability of the whole battery pack device.

### Third Embodiment

#### Retaining Member

[0076] A retaining member **30B** of a third embodiment of the present disclosure will be described below with reference to the drawings.

[0077] In the following description, description of the same components as those of the retaining members of the first and second embodiments will be omitted or simplified. In the retaining member **30B** of the third embodiment, a first cover portion and a second cover portion are respectively rotatably fixed to the same sidewall of a retaining portion, and the other sidewall of the retaining portion is provided with notches for accommodating the first cover portion.

[0078] FIG. **10** is a perspective view of the retaining member **30B** according to the third embodiment of the present disclosure, FIG. **11** is a schematic side view of the retaining member **30B** in an open state according to the third embodiment of the present disclosure, and FIG. **12** is a schematic side view of the retaining member **30B** in a closed state according to the third embodiment. As shown in the drawings, the retaining member **30B** includes the retaining portion **31** having a U-shaped cross section, and the first cover portion **32** and the second cover portion **33** rotatably fixed to the retaining portion **31**.

[0079] Specifically, in the retaining member **30B** of the present embodiment, the retaining portion **31** is composed of the first sidewall portion **311**, the second sidewall portion **312**, the bottom surface portion **313** connecting them, and the mounting portion **314** located at the end where the first sidewall portion **311** of the bottom surface portion **313** is located and extending downward from the bottom surface portion **313**. The first sidewall portion **311**, the second sidewall portion **312**, the bottom surface portion **313** and the mounting portion **314** are integrally formed and form a flat-bottomed U shape opened upward in a side view. The first sidewall portion **311** is formed by two parts separated at the left end and the right end, and two rotating shafts **315** extending from a left end and a right end to the center in a direction perpendicular to inner side surfaces thereof are formed at approximately half the height of the two parts, respectively. The two rotating shafts **315** extend to the center for a predetermined length to cooperate with the pivot hole portions **325** of the first cover portion **32** to be described below to mount the first cover portion **32**. An upper end of the first sidewall portion **311** is formed with two rotating shafts **316** extending from left and right ends to the center in a direction perpendicular to the inner side surfaces thereof, respectively, and the two rotating shafts **316** extend to the center by a predetermined length to cooperate with the pivot hole portions **335** of the second cover portion **33** to be described below to mount the second cover portion **33**. An upper end of the second sidewall portion **312** is formed with two notches **310** for accommodating side portions **326** of the first cover portion **32** to be described below. A protruding portion **312a** is formed on an outer side surface of the second sidewall portion **312** between the two notches **310**, and the protruding portion **312a** is used to be engaged with the engagement portion **334** of the second cover portion **33** to be described below.

[0080] The first cover portion **32** includes the flat plate-shaped cover plate portion **321**, the pressed portion **322** extending in a direction perpendicular to the plane of the cover plate portion **321** and formed at one end of the cover plate portion **321** in a front-rear direction, and the two side portions **326** extending in the direction perpendicular to the plane of the cover plate portion **321** and formed at the left end and the right end of the cover plate portion **321**, respectively. The two pivot hole portions **325** are formed at positions of the two side portions **326** close to the pressed portion **322**, respectively, the pivot hole portion **325** cooperates with the rotating shaft **315** of the retaining portion **31**, and the first cover portion **32** is rotatably mounted on the retaining portion **31** by sleeve-fitting the pivot hole portion **325** onto the rotating shaft **315**.

[0081] The second cover portion **33** includes the flat cover surface portion **331**, a sidewall portion **338** extending in a first direction perpendicular to a plane of the cover surface portion **331** and formed on one end of the cover surface portion **331** in the front-rear direction, and the two pivot hole portions **335** extending obliquely from the cover surface portion **331** and formed on the other end thereof. The pivot hole portion **335** is formed with a pivot hole that cooperates with the rotating shaft **316** of the retaining portion **31**, and the second cover portion **33** is rotatably mounted on the retaining portion **31** by sleeve-fitting the pivot hole onto the rotating shaft **316**. At a position slightly closer to the center than the pivot hole portion **335**, an extending portion **339** extending in the first direction is formed on the cover surface portion **331**. The sidewall portion **338** has a step portion **338a** further extending in an extending direction thereof. An outside of the sidewall portion **338** is formed with an engagement portion **333**, and an upper end of the engagement portion **333** extends to form an engaging portion **334** with a hook-shaped side surface in a side view. The second cover portion **33** may be fixed to the retaining portion **31** by engaging the engaging portion **334** with the protruding portion **312a** of the retaining portion **31**. In addition, at this time, the step portion **338a** of the sidewall portion **338** abuts on the side portions **326** of the first cover portion **32**, so that the first cover portion **32** is also fixed.

[0082] The first cover portion **32** and the second cover portion **33** are each rotatably mounted on the retaining portion **31**, and the retaining portion **31**, the first cover portion **32**, and the second cover portion **33** are sequentially arranged from bottom to top. As shown in FIG. **12**, when the first cover portion **32** is turned to the closed state, that is, the cover plate portion **321** is substantially parallel to the bottom surface portion **313** of the retaining portion **31**, surfaces of the side portions **326** and the bottom surface portion **313** of the retaining portion **31** facing each other, an inner side surface of the second sidewall portion **312** of the retaining portion **31**, and a surface of the pressed portion **322** of the first cover portion **32** facing the inner side surface constitute the first accommodating space **34** for accommodating one of the bus bars **20a** and **20b**. When the second cover portion **33** is also turned to the closed state, that is, when the cover surface portion **331** is substantially parallel to the cover plate portion **321** of the first cover portion **32**, surfaces of the cover surface portion **331** of the second cover portion **33** and the side portions **326** of the first cover portion **32** facing each other, an inner side surface of the sidewall portion **338** of the second cover portion **33**, and a surface of the extending portion **339** facing the inner side surface constitute the second accommodating space **35** for accommodating the other of the bus bars **20a** and **20b**.

[0083] According to the retaining member **30B** of the third embodiment of the present disclosure, the upper end of the second sidewall portion **312** of the retaining portion **31** is formed with the two notches **310**, and in the closed state, the side portions **326** of the first cover portion **32** are adapted and accommodated in the notches **310**. This mating structure not only provides physical locking, but also improves the stability of the first cover portion, and can reduce the risk of accidental loosening due to vibration or mechanical stress.

[0084] In addition, the inner side surface of the sidewall portion **338** of the second cover portion **33** and the extending portion **336** constitute the second accommodating space **35**, and displacement of the bus bars in the width direction is limited by the sidewall portion **332** and the extending portion **336** to ensure the stability of the bus bars. The structure effectively limits the unnecessary displacement of the bus bars in the width direction. Accordingly, potential damage caused by vibration or movement is reduced, and an additional guarantee is added to the stability and reliability of the whole battery pack device.

[0085] FIGS. **13A** to **13D** are schematic diagrams showing steps of installing the bus bars in the retaining member according to the third embodiment of the present disclosure. The steps of installing the bus bars **20a** and **20b** in the retaining member **30B** are described in detail below with reference to FIGS. **13A** to **13D**. It should be noted that the following description is based on the installing of two bus bars **20a** and **20b** stacked up and down in the retaining member **30B** as an example. For the installing of two bus bars **20a** and **20b** disposed side by side in the retaining

member **30B**, it is sufficient to rotate each member and the operating direction by **90** degrees in the following steps.

[0086] As shown in FIG. **13A**, in the retaining member **30B**, the first cover portion **32** and the second cover portion **33** are opened. At this point, the installer can start an installation process. The installer selects one bus bar (taking the bus bar **20a** as an example) and presses it from above into the retaining member **30B**, as indicated by an arrow. During this process, the bus bar **20a** comes into contact with and pushes the pressed portion **322** of the first cover portion **32** when pressed downward. This action triggers the first cover portion **32** to rotate downward about an axis (rotating shafts **315**) into the closed state, as shown in FIG. **13B**. As the bus bar **20a** continues moving until the bus bar **20a** comes into contact with the bottom surface portion **313** of the retaining portion **31**, the pressed portion **322** of the first cover portion **32** is pushed to be substantially perpendicular to the bottom surface portion **313** of the retaining portion **31**. At this time, the first cover portion **32** is completely closed and covers the bus bar **20a**, and the bus bar **20a** is stably accommodated in the first accommodating space **34**, thereby completing installation of the bus bar **20a**.

[0087] Then, as shown in FIG. **13C**, the installer places the second bus bar (bus bar **20b**) from above into the second accommodating space **35** in a similar way. In the final step, as shown in FIG. **13D**, the installer pushes the second cover portion **33** so as to rotate the second cover portion **33** to the closed state and presses the second cover portion **33** downward, so that the engaging portion **334** is engaged with the protruding portion **312a** of the retaining portion **31**, thereby fixing the second cover portion **33** to the retaining portion **31**. At this time, the step portion **338a** of the sidewall portion **338** abuts on the side portion **326** of the first cover portion **32**, so that the first cover portion **32** is also fixed. In this manner, the installation of the bus bars **20a** and **20b** is completed.

[0088] Similar to the retaining members of the first and second embodiments, the retaining member **30B** in the third embodiment of the present disclosure allows the two bus bars **20a** and **20b** to be fixed in the same retaining member, so that space can be efficiently utilized in the compact arrangement of the battery modules, and a space layout in the battery pack can be optimized. Moreover, by using the design of the pressed portion **322**, the first cover portion **32** is automatically pushed to rotate and close when installing the first bus bar, the number of parts is reduced, and the operation steps of installing the bus bars are also simplified.

[0089] In addition, in the third embodiment of the present disclosure, the design of the retaining member **30B** allows the installer to engage the retaining portion **31** and fix the first cover portion **32** and the second cover portion **33** together by pressing only the second cover portion **33** without pressing the first cover portion **32** and the second cover portion **33** in sequence. This design further simplifies the operation steps, improves installation efficiency, and makes it easier for the bus bars **20** to be fixed in the retaining member **30B**. Moreover, since the first cover portion **32** and the second cover portion **33** are opened toward the same side and an opening degree is relatively large, the operation space is larger and it is easier to visually recognize the bus bars **20** when the bus bars **20** are installed.

[0090] In the closed state, the side portions **326** of the first cover portion **32** are adapted and accommodated in the notches **310**, and this mating structure not only provides physical locking, but also improves the stability of the first cover portion, and can reduce the risk of accidental loosening due to vibration or mechanical stress.

[0091] In addition, the inner side surface of the sidewall portion **338** of the second cover portion **33** and the extending portion **336** constitute the second accommodating space **35**, and the sidewall portion **332** and the extending portion **336** limit displacement of the bus bars in the width direction and thus effectively limit the unnecessary displacement of the bus bars in the width direction. Accordingly, potential damage caused by vibration or movement is reduced, and an additional guarantee is added to the stability and reliability of the whole battery pack device.

[0092] The present disclosure is not limited to the above-described embodiments, and

modifications, improvements, or the like can be made as appropriate. For example, in the above-described embodiments, the two bus bars stacked up and down are fixed by one retaining member, and the two bus bars disposed side by side are fixed by three retaining members, but the present disclosure is not limited thereto. The number of retaining members may be flexibly increased or decreased according to the size of an installation space, the number of bus bars and the complexity of electrical connection. For example, for electronic devices or battery modules requiring high-density wiring, a more compact retaining member layout scheme may be designed to optimize space usage efficiency. In addition, it is also possible to consider using a retaining member in a mountain shape (or other geometric shapes) to fix the bus bars disposed side by side.

[0093] In addition, in the above-described embodiments, the rotating shafts are provided on the retaining portion in the retaining member, and the pivot holes cooperating therewith are disposed on the first and second cover portions, respectively. The present disclosure may also adopt a reverse configuration, that is, a pivot hole is provided on the retaining portion, and the rotating shafts are disposed on the first and second cover portions, respectively. The adjustability of this design provides greater flexibility for assembly and maintenance of the product, while also optimizing the design of a rotating mechanism according to specific application requirements.

[0094] In addition, the fixing of the first cover portion and the second cover portion to the retaining portion is not limited to the engagement mode of the engaging portion and the slot, and other fixing mechanisms may also be adopted, such as using structures such as protruding stripes and quick locks.

[0095] According to the battery device in the present disclosure, the two bus bars can be fixed by using one retaining member, the requirement of two or more independent retaining members in the past is eliminated, the number of parts is reduced, and operation steps of installing the bus bars are simplified. Moreover, the space can be efficiently utilized in the compact arrangement of the battery modules, the space layout in the battery device is optimized, a more efficient, compact and economical battery device for an electric vehicle can be manufactured, and important support is provided for popularization and development of the electric vehicle.

[0096] In addition, the present disclosure includes at least the following items, corresponding components or the like in the above-described embodiments are indicated in brackets, but the present disclosure is not limited thereto.

#### Scheme 1

[0097] A battery device for an electric vehicle, including: [0098] a plurality of battery modules (battery modules **11**); and [0099] a housing (housing **12**) configured to accommodate the plurality of battery modules, wherein [0100] the housing is configured to accommodate a pair of strip-shaped bus bars (bus bars **20a** and **20b**) and a retaining member (retaining member **30**) retaining the pair of bus bars, [0101] the retaining member includes [0102] a retaining portion (retaining portion **31**) with one side open and having a U-shaped cross section, and [0103] a first cover portion (first cover portion **32**) and a second cover portion (second cover portion **33**) that are rotatably disposed on the retaining portion, [0104] a first bus bar of the pair of bus bars is accommodated between the retaining portion and the first cover portion, [0105] a second bus bar of the pair of bus bars is accommodated between the first cover portion and the second cover portion, and [0106] a pressed portion (pressed portion **322**) is formed on the first cover portion, and is formed to be pressed and cause the first cover portion to rotate when the first bus bar is placed between the retaining portion and the first cover portion.

[0107] By adopting the battery device of scheme 1, the two bus bars can be fixed by using one retaining member, thereby eliminating the need for two or more independent retaining members in the past. Moreover, by adopting the design of the pressed portion, the first cover portion is automatically pushed to rotate and close when installing the first bus bar. This design not only reduces the number of parts, but also simplifies the operation steps of installing the bus bars. In this way, the present disclosure not only improves assembly efficiency, but also enhances overall

structural stability and a space utilization rate of the battery pack.

#### Scheme 2

[0108] In the battery device according to scheme 1, [0109] the retaining member includes a first rotating shaft (rotating shaft **315**) configured to rotate the first cover portion and a second rotating shaft (rotating shaft **316**) configured to rotate the second cover portion, [0110] the retaining portion has facing sidewalls (first sidewall portion **311** and second sidewall portion **312**), [0111] the first rotating shaft is disposed on one of the facing sidewalls (second sidewall portion **312**), [0112] the second rotating shaft is disposed on the other of the facing sidewalls (the second sidewall portion **312**), [0113] a first engagement portion (engagement portion **324**) engaged with the retaining portion is formed on the first cover portion, and [0114] a second engagement portion (engagement portion **333**) engaged with the retaining portion is formed on the second cover portion.

[0115] By adopting the battery device of scheme 2, the first cover portion and the second cover portion are respectively disposed on different sidewall portions, and the first cover portion is automatically pushed to rotate around the first rotating shaft and close when installing the first bus bar. At this time, the first engagement portion can be operated to fix the first cover portion to the retaining portion, thereby simplifying the operation steps of installing the bus bars.

#### Scheme 3

[0116] In the battery device according to scheme 1, [0117] the retaining member (retaining member **30A**) includes a first rotating shaft (rotating shaft **315**) configured to rotate the first cover portion and a second rotating shaft (rotating shaft **316**) configured to rotate the second cover portion, [0118] the retaining portion has facing sidewalls (first sidewall portion **311** and second sidewall portion **312**), [0119] both the first rotating shaft and the second rotating shaft are provided on one of the facing sidewalls (first sidewall portion **311**), [0120] the first cover portion has a contact wall (contact wall **323**) configured to be in contact with the second cover portion, and [0121] an engagement portion (engagement portion **333**) is formed on the second cover portion, and the engagement portion is formed to be engaged with the retaining portion in a state in which the second cover portion is in contact with the contact wall.

[0122] By adopting the battery device of scheme 3, the engagement portion is formed to be engaged with the retaining portion in the state in which the second cover portion is in contact with the contact wall. This design allows an installer to engage the retaining portion and fix the first cover portion and the second cover portion together by pressing only the second cover portion without pressing the first cover portion and the second cover portion in sequence, operation steps are further simplified, and installation efficiency is improved.

#### Scheme 4

[0123] In the battery device according to scheme 3, [0124] a groove portion (groove **337**) configured to accommodate a portion of the second bus bar is formed in the second cover portion, and [0125] the second bus bar is retained between one side surface of the contact wall of the first cover portion and one side surface of the groove portion of the second cover portion.

[0126] By adopting the battery device of scheme 4, the second bus bar is retained between the contact wall of the first cover portion and the groove portion of the second cover portion, the groove portion and the contact wall limit displacement of the bus bars in the width direction and thus effectively limit the unnecessary displacement of the bus bars. Accordingly, the potential damage caused by vibration or movement is reduced, and an additional guarantee is added to the stability and reliability of the whole battery pack device.

#### Scheme 5

[0127] In the battery device according to scheme 1, [0128] the retaining member (retaining member **30B**) includes a first rotating shaft (rotating shaft **315**) configured to rotate the first cover portion and a second rotating shaft (rotating shaft **316**) configured to rotate the second cover portion, [0129] the retaining portion has facing sidewalls (first sidewall portion **311** and second sidewall portion **312**), [0130] both the first rotating shaft and the second rotating shaft are provided

on one of the facing sidewalls (first sidewall portion **311**), [0131] a contact portion configured to be in contact with the second cover portion is provided on the other of the facing sidewalls, [0132] the contact portion is allowed to be inserted by a part of the first cover portion, and [0133] an engagement portion (protruding portion **312a**) is formed on the contact portion, and the engagement portion is formed to be engaged with the second cover portion in a state in which the contact portion is in contact with the second cover portion.

[0134] With the battery device of scheme 5, a contact portion configured to be in contact with the second cover portion is provided on the other of the facing sidewalls, the contact portion is allowed to be inserted by the part of the first cover portion, and the engagement portion is formed to be engaged with the second cover portion in a state in which the contact portion is in contact with the second cover portion. The engagement structure not only provides physical locking, but also improves stability of the first cover portion, and can reduce a risk of accidental loosening due to vibration or mechanical stress.

#### Scheme 6

[0135] In the battery device according to scheme 5, [0136] the second cover portion is provided with a protruding portion (extending portion **339**) protruding from an inner side surface, and [0137] the second bus bar is retained between one side surface of the contact portion and one side surface of the protruding portion.

[0138] By adopting the battery device of scheme 6, the contact portion and the protruding portion form a space for accommodating the second bus bar, limit the displacement of the second bus bar and thus effectively limit the unnecessary displacement of the bus bars. Accordingly, the potential damage caused by vibration or movement is reduced, and the additional guarantee is added to the stability and reliability of the whole battery pack device.

#### Scheme 7

[0139] In the battery device according to any one of schemes 2, 3 and 5, [0140] the first bus bar is retained between the pressed portion of the first cover portion and a sidewall of the facing sidewalls of the retaining portion facing the pressed portion.

[0141] By adopting the battery device of scheme 7, the pressed portion and the sidewall of the retaining portion facing the pressed portion form a space for accommodating the second bus bar, limit the displacement of the second bus bar, and thus effectively limit the unnecessary displacement of the bus bars. Accordingly, the potential damage caused by vibration or movement is reduced, and the additional guarantee is added to the stability and reliability of the whole battery pack device.

#### Scheme 8

[0142] In the battery device according to scheme 1, [0143] the pair of bus bars is retained by the retaining member in a state in which thickness directions face the same direction.

[0144] By adopting the battery device of scheme 8, the pair of bus bars is retained by the retaining member in the state in which the thickness directions face the same direction, that is, the pair of bus bars is allowed to be stacked up and down or left and right side by side, a space can be efficiently utilized in the compact arrangement of the battery modules, and a space layout in the battery pack can be optimized.

#### Scheme 9

[0145] A method for assembling bus bars, the bus bars being assembled in a battery device using a retaining member, the battery device being a battery device for an electric vehicle and including a plurality of battery modules (battery modules **11**), and a housing (housing **12**) configured to accommodate the plurality of battery modules, the pair of strip-shaped bus bars (bus bars **20a** and **20b**) and the retaining member (retaining member **30**) retaining the pair of bus bars being accommodated in the housing, the retaining member including a retaining portion (retaining portion **31**) with one side open and having a U-shaped cross section, and a first cover portion (first cover portion **32**) and a second cover portion (second cover portion **33**) rotatably disposed on the

retaining portion, and a pressed portion (pressed portion **322**) being formed on the first cover portion, the method comprising: [0146] pressing a first bus bar (bus bar **20a**) of the pair of bus bars toward the pressed portion to rotate the first cover portion, and accommodating the first bus bar between the retaining portion and the first cover portion; [0147] disposing a second bus bar (bus bar **20b**) of the pair of bus bars on the rotated first cover portion; and [0148] rotating the second cover portion to accommodate the second bus bar between the first cover portion and the second cover portion.

[0149] By adopting the method for assembling bus bars of scheme 9, the two bus bars can be fixed by using one retaining member in the battery device, thereby eliminating the need for two or more independent retaining members in the past. Further, by adopting the design of the pressed portion, the first cover portion is automatically pushed to rotate and close when installing the first bus bar, and the operation steps of installing the bus bars are simplified. Not only is assembly efficiency improved, but also overall structural stability and a space utilization rate of the battery pack are enhanced.

#### Scheme 10

[0150] In the method for assembling bus bars according to scheme 9, [0151] after the first bus bar is accommodated between the retaining portion and the first cover portion, a first engagement portion (engagement portion **324**) formed on the first cover portion is engaged with the retaining portion, and [0152] after the second bus bar is accommodated between the first cover portion and the second cover portion, a second engagement portion (engagement portion **333**) formed on the second cover portion is engaged with the retaining portion.

[0153] By adopting the bus bar assembly method of scheme 10, the first cover portion is automatically pushed to rotate and close when installing the first bus bar. At this time, the first engagement portion can be operated to fix the first cover portion to the retaining portion, then the second bus bar is installed, and finally, the second engagement portion is operated to fix the second cover portion to the retaining portion, thereby simplifying the operation steps of installing the bus bars.

#### Scheme 11

[0154] In the method for assembling bus bars according to scheme 9, [0155] after the second bus bar is accommodated between the first cover portion and the second cover portion, an engagement portion (engagement portion **333**) formed on the second cover portion is engaged with the retaining portion in a state in which the second cover portion is in contact with a contact wall (contact wall **323**) of the first cover portion.

[0156] By adopting the bus bar assembly method of scheme 11, the retaining portion is engaged and the first cover portion and the second cover portion are fixed together by pressing only the second cover portion without pressing the first cover portion and the second cover portion in sequence, operation steps are further simplified, and installation efficiency is improved.

## Claims

1. A battery device for an electric vehicle, comprising: a plurality of battery modules; and a housing configured to accommodate the plurality of battery modules, wherein the housing configured to accommodate a pair of strip-shaped bus bars and a retaining member retaining the pair of bus bars, the retaining member includes a retaining portion with one side open and having a U-shaped cross section, and a first cover portion and a second cover portion that are rotatably disposed on the retaining portion, a first bus bar of the pair of bus bars is accommodated between the retaining portion and the first cover portion, a second bus bar of the pair of bus bars is accommodated between the first cover portion and the second cover portion, and a pressed portion is formed on the first cover portion, and is formed to be pressed and cause the first cover portion to rotate when the first bus bar is placed between the retaining portion and the first cover portion.

**2.** The battery device according to claim 1, wherein the retaining member includes a first rotating shaft configured to rotate the first cover portion and a second rotating shaft configured to rotate the second cover portion, the retaining portion has facing sidewalls, the first rotating shaft is disposed on one of the facing sidewalls, the second rotating shaft is disposed on the other of the facing sidewalls, a first engagement portion engaged with the retaining portion is formed on the first cover portion, and a second engagement portion engaged with the retaining portion is formed on the second cover portion.

**3.** The battery device according to claim 1, the retaining member includes a first rotating shaft configured to rotate the first cover portion and a second rotating shaft configured to rotate the second cover portion, the retaining portion has facing sidewalls, both the first rotating shaft and the second rotating shaft are provided on one of the facing sidewalls, the first cover portion has a contact wall configured to be in contact with the second cover portion, and an engagement portion is formed on the second cover portion, and the engagement portion is formed to be engaged with the retaining portion in a state in which the second cover portion is in contact with the contact wall.

**4.** The battery device according to claim 3, a groove portion configured to accommodate a portion of the second bus bar is formed in the second cover portion, and the second bus bar is retained between one side surface of the contact wall of the first cover portion and one side surface of the groove portion of the second cover portion.

**5.** The battery device according to claim 1, wherein the retaining member includes a first rotating shaft configured to rotate the first cover portion and a second rotating shaft configured to rotate the second cover portion, the retaining portion has facing sidewalls, both the first rotating shaft and the second rotating shaft are provided on one of the facing sidewalls, a contact portion configured to be in contact with the second cover portion is provided on the other of the facing sidewalls, the contact portion is allowed to be inserted by a part of the first cover portion, and an engagement portion is formed on the contact portion, and the engagement portion is formed to be engaged with the second cover portion in a state in which the contact portion is in contact with the second cover portion.

**6.** The battery device according to claim 5, wherein the second cover portion is provided with a protruding portion protruding from an inner side surface, and the second bus bar is retained between one side surface of the contact portion and one side surface of the protruding portion.

**7.** The battery device according to claim 2, wherein the first bus bar is retained between the pressed portion of the first cover portion and a sidewall of the facing sidewalls of the retaining portion facing the pressed portion.

**8.** The battery device according to claim 1, wherein the pair of bus bars is retained by the retaining member in a state in thickness directions face the same direction.

**9.** A method for assembling bus bars in a battery device for an electric vehicle using a retaining member, wherein the battery device includes a plurality of battery modules and a housing configured to accommodate the plurality of battery modules, the housing is configured to accommodate the pair of strip-shaped bus bars and the retaining member retaining the pair of bus bars, the retaining member includes a retaining portion with one side open and having a U-shaped cross section, and a first cover portion and a second cover portion that are rotatably disposed on the retaining portion, a pressed portion is formed on the first cover portion, and the method comprises: pressing a first bus bar of the pair of bus bars toward the pressed portion to rotate the first cover portion, and accommodating the first bus bar between the retaining portion and the first cover portion; disposing a second bus bar of the pair of bus bars on the rotated first cover portion; and rotating the second cover portion to accommodate the second bus bar between the first cover portion and the second cover portion.

**10.** The method for assembling bus bars according to claim 9, wherein after the first bus bar is accommodated between the retaining portion and the first cover portion, a first engagement portion formed on the first cover portion is engaged with the retaining portion, and after the second bus bar is accommodated between the first cover portion and the second cover portion, a second

engagement portion formed on the second cover portion is engaged with the retaining portion.

**11.** The method for assembling bus bars according to claim 9, wherein after the second bus bar is accommodated between the first cover portion and the second cover portion, an engagement portion formed on the second cover portion is engaged with the retaining portion in a state in which the second cover portion is in contact with a contact wall of the first cover portion.

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